Cartridge connections

J6 CARTRIDGE_F

GND	1	31	VCC		
CPU_A11	2	32	M2		
CPU_A10	3	33	CPU_A12		
CPU_A9	4	34	CPU_A13		
CPU_A8	5	35	CPU_A14		
CPU_A7	6	36	CPU_D7		
CPU_A6	7	37	CPU_D6		
CPU_A5	8	38	CPU_D5		
CPU_A4	9	39	CPU_D4		
CPU_A3	10	40	CPU_D3		
CPU_A2	11	41	CPU_D2		
CPU_A1	12	42	CPU_D1		
CPU_A0	13	43	CPU_D0		
CPU_RW	14	44	ROMSEL		
IRQ	15	45	AUD_2A03		
GND	16	46	AUD_RF		
PPU_RD	17	47	PPU_WR		
CIRAM_A10	18	48	CIRAM_CE		
PPU_A6	19	49	PPU_A13		
PPU_A5	20	50	PPU_A7		
PPU_A4	21	51	PPU_A8		
PPU_A3	22	52	PPU_A9		
PPU_A2	23	53	PPU_A10		
PPU_A1	24	54	PPU_A11		
PPU_A0	25	55	PPU_A12		
PPU_D0	26	56	PPU_A13		
PPU_D1	27	57	PPU_D7		
PPU_D2	28	58	PPU_D6		
PPU_D3	29	59	PPU_D5		
VCC	30	60	PPU_D4		

J5 CONSOLE_M

GND	1	31	VCC		
CPU_A11	2	32	M2		
CPU_A10	3	33	CPU_A12		
CPU_A9	4	34	CPU_A13		
CPU_A8	5	35	CPU_A14		
CPU_A7	6	36	CPU_D7		
CPU_A6	7	37	CPU_D6		
CPU_A5	8	38	CPU_D5		
CPU_A4	9	39	CPU_D4		
CPU_A3	10	40	CPU_D3		
CPU_A2	11	41	CPU_D2		
CPU_A1	12	42	CPU_D1		
CPU_A0	13	43	CPU_D0		
CPU_RW	14	44	ROMSEL		
IRQ	15	45	AUD_2A03		
GND	16	46	AUD_EPSM		
PPU_RD	17	47	PPU_WR		
CIRAM_A10	18	48	CIRAM_CE		
PPU_A6	19	49	PPU_A13		
PPU_A5	20	50	PPU_A7		
PPU_A4	21	51	PPU_A8		
PPU_A3	22	52	PPU_A9		
PPU_A2	23	53	PPU_A10		
PPU_A1	24	54	PPU_A11		
PPU_A0	25	55	PPU_A12		
PPU_D0	26	56	PPU_A13		
PPU_D1	27	57	PPU_D7		
PPU_D2	28	58	PPU_D6		
PPU_D3	29	59	PPU_D5		
VCC	30	60	PPU_D4		

Bus buffers

U4 74HCT541

CPU_D7	2	18	B_CPU_D7
CPU_D6	3	17	B_CPU_D6
CPU_D5	4	16	B_CPU_D5
CPU_D4	5	15	B_CPU_D4
CPU_D3	6	14	B_CPU_D3
CPU_D2	7	13	B_CPU_D2
CPU_D1	8	12	B_CPU_D1
CPU_D0	9	11	B_CPU_D0
GND	1		
GND	19		

U2 74HCT541

M2	2	18	B_M2
CPU_A11	3	17	B_CPU_A11
CPU_A12	4	16	B_CPU_A12
CPU_A10	5	15	B_CPU_A10
CPU_A13	6	14	B_CPU_A13
CPU_A9	7	13	B_CPU_A9
CPU_A14	8	12	B_CPU_A14
CPU_A8	9	11	B_CPU_A8
GND	1		
GND	19		

U5 74HCT541

CPU_RW	2	18	B_CPU_RW
ROMSEL	3	17	B_ROMSEL
GND	4	16	
GND	5	15	
GND	6	14	
GND	7	13	
GND	8	12	
GND	9	11	
GND	1		
GND	19		

U3 74HCT541

CPU_A7	2	18	B_CPU_A7
CPU_A6	3	17	B_CPU_A6
CPU_A5	4	16	B_CPU_A5
CPU_A4	5	15	B_CPU_A4
CPU_A3	6	14	B_CPU_A3
CPU_A2	7	13	B_CPU_A2
CPU_A1	8	12	B_CPU_A1
CPU_A0	9	11	B_CPU_A0
GND	1		
GND	19		

C7 0.1uF
C5 0.1uF
C8 0.1uF
C6 0.1uF

Address and data daughterboard connectors

J4 EPSM_ADDR

GND	1	2	VCC
B_CPU_A11	3	4	B_M2
B_CPU_A10	5	6	B_CPU_A12
B_CPU_A9	7	8	B_CPU_A13
B_CPU_A8	9	10	B_CPU_A14
B_CPU_A7	11	12	
B_CPU_A6	13	14	EPSM_A1
B_CPU_A5	15	16	EPSM_A0
B_CPU_A4	17	18	EPSM_CE3
B_CPU_A3	19	20	EPSM_CE2
B_CPU_A2	21	22	EPSM_CE1
B_CPU_A1	23	24	
B_CPU_A0	25	26	GND
B_CPU_RW	27	28	B_ROMSEL
	29	30	GND

J1 EPSM_DATA

GND	1	2	VCC
YM_CS	3	4	B_CPU_D7
YM_A0	5	6	B_CPU_D6
YM_A1	7	8	B_CPU_D5
YM_D7	9	10	B_CPU_D4
YM_D6	11	12	B_CPU_D3
YM_D5	13	14	B_CPU_D2
YM_D4	15	16	B_CPU_D1
YM_D3	17	18	B_CPU_D0
YM_D2	19	20	EPSM_A1
YM_D1	21	22	EPSM_A0
YM_D0	23	24	EPSM_CE3
GND	25	26	EPSM_CE2
GND	27	28	EPSM_CE1
VCC	29	30	GND

Audio daughterboard connector and jack

J2 EPSM_AUDIO

CLK_8MHz	2	1	VCC
GND	4	3	AUD_RF
YM_LRCK	6	5	GND
YM_SDAT	8	7	EPSM_SND_R
YM_BCLK	10	9	GND
GND	12	11	EPSM_SND_L

J3 EPSM_SND_OUT
GND, S, EPSM_SND_R, R2, EPSM_SND_L, R1, R1N, T, TN
R3 1K AUD_EPSM

YMF288

U1 YMF288

YM_D0	2	26	YM_LRCK
YM_D1	3	25	
YM_D2	4	24	YM_SDAT
YM_D3	5	23	YM_BCLK
YM_D4	6	22	
YM_D5	7	15	D2 IRQ
YM_D6	8		CD4148WSP
YM_D7	9		
YM_A0	20	11	RESET
YM_A1	21	12	VCC
YM_CS	17	10	
GND	18	13	
VCC	19	28	CLK_8MHz
		16	GND

C2 0.1uF
R1 330
D1 LED

Synchronized RESET signal

U7 74LVC1GU04
U6 74LVC1G175
C11 0.1uF
C3 0.1uF
C4 5n
D3 CD4148WSP
R2 1K
JP1
If not using sync RESET, unpopulate D flip flop, break jumper and short M2_HALT to RESET

Crystal Oscillator

X1 SG-8002CA
C1 68pF

C10 100uF
C9 100uF

Some pinouts subject to change over time for ease of layout
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Sheet: /
File: EPSM YMF288 Modular Famicom Horizontal.kicad_sch

Title: EPSM – Modular Famicom H Connector

Size: A4	Date: 2026-10-30	**Rev: r0.1.0**
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